

Movie Ticket Booking System

A Command-Line Booking System Implemented in C

Project Report

CSC 1031 - Software Development Fundamentals

University of Sri Jayewardenepura



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Submitted Date: [24/07/2026]

Introduction

Cinemas that rely on a paper diary to track bookings run into the same problems every weekend: double booked seats, miscounted takings, and no quick way to check whether a seat is still free. The Movie Ticket Booking System is a command line C application that removes that friction, letting a small cinema manage seat reservations, pricing, and revenue for a fixed set of movies and showtimes entirely in memory, with no database or external dependency.

The project was built with the standard C library only (no external dependencies) and compiles with a single `gcc *.c -o booking` command.

The system is menu-driven: on start-up it builds a fixed catalogue of three movies and their showtimes (The system models three movies, each with two showtimes, and every showtime has its own 5×10 seat grid.), then loops on a role menu (Customer / Admin / Exit). Customers can view showtimes, view seat maps, book seats, and cancel bookings; Admins get those features plus search and a revenue report, gated behind a login with a limited number of attempts.

Team Contribution Summary

The table below lists each member, their assigned module(s), and a short description of their specific contribution to the project.

Member	Module(s)	Contribution
T.S.P.Fernando	Auth,c/auth.h, config.h, README.md, main.c	Project planning, repository setup, backend development
A.J.M.H.M.Aberathne	pricing.c / pricing.h	Frontend development and UI design
H.D.J.Kaveesha	movie.c/movie.h	Database design and implementation
M.P.G.H.Pathirana	display.c / display.h, seat.c/seat.h	Testing, debugging, and documentation
K.T.D.K.H.Wijethunga	booking.c/ booking.h, input.c/input.h	booking/input files

Design Overview

1. Data Structures

- Seat (seat.h): the smallest stateful object in the system - an isBooked flag, customerName, pricePaid, and isStudent/isSenior flags recording which discount applied to that specific seat.
- Showtime (movie.h): owns a full NUM_ROWS × SEATS_PER_ROW grid of Seat, plus a time label and running totals ticketsSold and totalRevenue.
- Movie (movie.h): a title paired with a fixed-size array of Showtime. initMovies() builds the hardcoded catalogue of three movies.

2. Pricing & Discount Logic

Seat Tier	Rows	Base Price
Regular	A – B	Rs. 500.00
Premium	C – D	Rs. 750.00
VIP	E	Rs. 1000.00

- Student: 10% off base price. Senior citizen: 20% off base price. Only one of these two categories can apply per booking.
- Group: an additional 10% off when 4 or more seats are booked in a single transaction; this combines additively with the category discount.

Final price = $\text{basePrice} \times (1 - (\text{category\%} + \text{group\%}) / 100)$. Example: a Premium seat (Rs. 750) booked by a student as part of a 5 seat group booking gets 10% + 10% = 20% off, giving $750 \times 0.80 = \text{Rs. } 600.00$. On cancellation, cancelSeat() subtracts the seat's stored pricePaid from totalRevenue and decrements ticketsSold before clearing the seat, so refunds are never left counted in the revenue total.

3. Module Connections

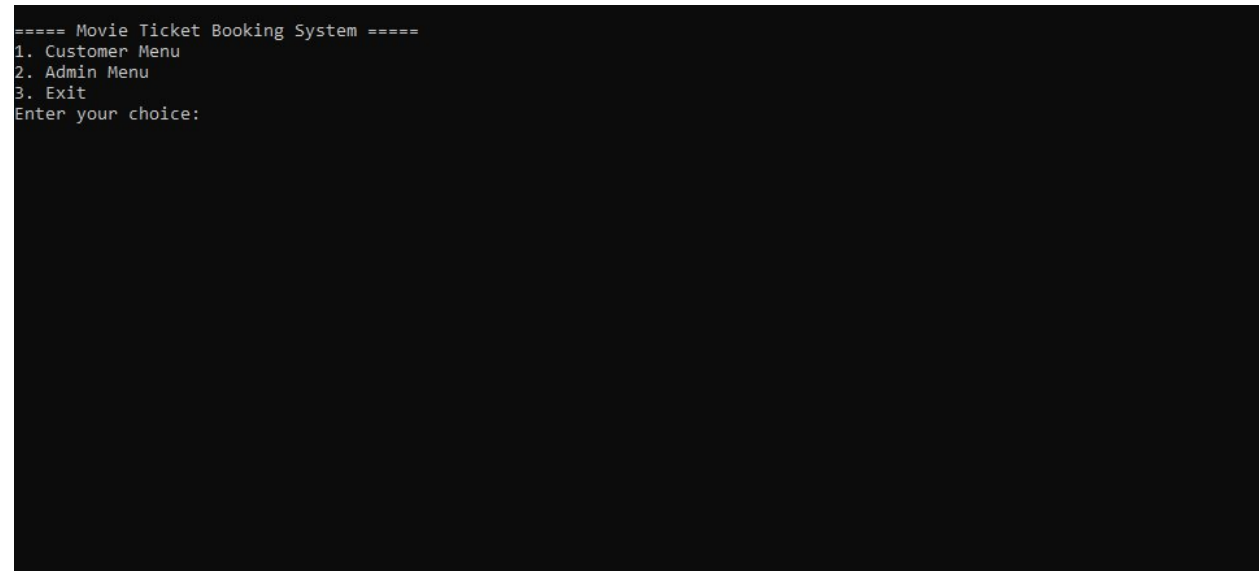
The list below shows how the source files call into one another, from the top-level menu loop down to the lowest-level lookups.

- main.c - role/menu loops; wires user choices to booking, display, and input.
- main.c calls readInt() / readLine() / readChar() in input.c for all validated input.
- main.c calls authenticateAdmin() in auth.c to gate the Admin Menu.
- main.c calls printShowtimesList(), printSeatMap(), printRevenueReport() in display.c for all screen output.
- main.c calls bookSeat(), cancelSeat(), searchByName(), searchBySeat() in booking.c for all booking operations.
- booking.c calls calculateSeatPrice() in pricing.c to obtain the final price for a seat.
- booking.c and display.c both call seatTierForRow(), seatTierSymbol(), isValidSeatPosition() in seat.c for tier lookups and position validation.
- movie.c's initMovies() builds the Movie / Showtime / Seat data that every module above operates on.

Screenshots & Program Walkthrough

This section walks through a single continuous run of the compiled program, from launch to exit, with a placeholder for each screenshot that should be captured directly from the terminal or IDE.

Screenshot 1: The main menu — the role menu shown at start-up.

A screenshot of a terminal window showing the main menu of a 'Movie Ticket Booking System'. The text is white on a black background. The menu lists three options: '1. Customer Menu', '2. Admin Menu', and '3. Exit'. Below the list, it prompts the user to 'Enter your choice:'.

```
==== Movie Ticket Booking System ====  
1. Customer Menu  
2. Admin Menu  
3. Exit  
Enter your choice:
```

Screenshot 2: The seat map before any bookings are made.

```
===== Movie Ticket Booking System =====
1. Customer Menu
2. Admin Menu
3. Exit
Enter your choice: 1

----- Customer Menu -----
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Back to Main Menu
Enter your choice: 2

--- Movies & Showtimes ---
1. Ben 10 Alien Force
   1) 10:00 AM
   2) 6:30 PM
2. Odyssey
   1) 1:00 PM
   2) 9:00 PM
3. Spider Man 4
   1) 11:30 AM
   2) 4:15 PM
Select movie number (0 to cancel): 2
Select showtime number (0 to cancel): 1

Seat map for 1:00 PM    (. = free, X = booked)
   1  2  3  4  5  6  7  8  9 10
A [R]  .  .  .  .  .  .  .  .  .  .
B [R]  .  .  .  .  .  .  .  .  .  .
C [P]  .  .  .  .  .  .  .  .  .  .
D [P]  .  .  .  .  .  .  .  .  .  .
E [V]  .  .  .  .  .  .  .  .  .  .
Legend: [R]=Regular Rs.500  [P]=Premium Rs.750  [V]=VIP Rs.1000

----- Customer Menu -----
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Back to Main Menu
Enter your choice:
```

```
Seat map for 10:00 AM    (. = free, X = booked)
   1  2  3  4  5  6  7  8  9 10
A [R]  .  .  .  .  .  .  .  .  .  .
B [R]  .  .  .  .  .  .  .  .  .  .
C [P]  .  .  .  .  .  .  .  .  .  .
D [P]  .  .  .  .  .  .  .  .  .  .
E [V]  .  .  .  .  .  .  .  .  .  .
Legend: [R]=Regular Rs.500  [P]=Premium Rs.750  [V]=VIP Rs.1000
```

Screenshot 3: A successful seat booking, showing the calculated price.

```
===== Movie Ticket Booking System =====
1. Customer Menu
2. Admin Menu
3. Exit
Enter your choice: 1

----- Customer Menu -----
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Back to Main Menu
Enter your choice: 3

--- Movies & Showtimes ---
1. Ben 10 Alien Force
   1) 10:00 AM
   2) 6:30 PM
2. Odyssey
   1) 1:00 PM
   2) 9:00 PM
3. Spider Man 4
   1) 11:30 AM
   2) 4:15 PM
Select movie number (0 to cancel): 3
You selected: Spider Man 4, Select showtime number (0 to cancel): 2

Seat map for 4:15 PM   (. = free, X = booked)
   1  2  3  4  5  6  7  8  9 10
A [R]  .  .  .  .  .  .  .  .  .  .
B [R]  .  .  .  .  .  .  .  .  .  .
C [P]  .  .  .  .  .  .  .  .  .  .
D [P]  .  .  .  .  .  .  .  .  .  .
E [V]  .  .  .  .  .  .  .  .  .  .
Legend: [R]=Regular Rs.500  [P]=Premium Rs.750  [V]=VIP Rs.1000
How many seats to book in this transaction? 3
Customer name: Amal
Discount category:
  0. None
  1. Student (10% off)
  2. Senior Citizen (20% off)
Choose category: 1
```

```
Seat 1 of 3:
  Row letter (A-E): D
  Seat number (1-10): 4
  Booked seat D4 for Rs. 675.00

Seat 2 of 3:
  Row letter (A-E): D
  Seat number (1-10): 5
  Booked seat D5 for Rs. 675.00

Seat 3 of 3:
  Row letter (A-E): E
  Seat number (1-10): 5
  Booked seat E5 for Rs. 900.00

You selected these seats: D4, D5, E5
Total price = Rs. 2250.00
Confirm this booking? (y/n): Y

3 of 3 requested seat(s) booked successfully.
```

Screenshot 4: An attempt to book an already-taken seat, showing the error message (no crash).

```
27/11/2021
Select movie number (0 to cancel): 3
You selected: Spider Man 4, Select showtime number (0 to cancel): 2

Seat map for 4:15 PM  (. = free, X = booked)
  1  2  3  4  5  6  7  8  9 10
A [R] . . . . . X X . .
B [R] . . . . . . . . .
C [P] . . . . . . . . .
D [P] . . . X X . . . .
E [V] . . . . X . . . .
Legend: [R]=Regular Rs.500  [P]=Premium Rs.750  [V]=VIP Rs.1000
How many seats to book in this transaction? 2
Customer name: Daham
Discount category:
  0. None
  1. Student (10% off)
  2. Senior Citizen (20% off)
Choose category: 0

Seat 1 of 2:
  Row letter (A-E): D
  Seat number (1-10): 5
  Error: seat D5 is already booked.
  Book again with a different seat? (y/n): Y
  Row letter (A-E): C
  Seat number (1-10): 6
  Booked seat C6 for Rs. 750.00

Seat 2 of 2:
  Row letter (A-E): C
  Seat number (1-10): 11
Invalid input. Please enter a whole number between 1 and 10.
  Seat number (1-10): 7
  Booked seat C7 for Rs. 750.00

You selected these seats: C6, C7
Total price = Rs. 1500.00
Confirm this booking? (y/n): y

2 of 2 requested seat(s) booked successfully.
```

Screenshot 5: A booking cancellation, showing the seat becoming free again.

```

----- Customer Menu -----
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Back to Main Menu
Enter your choice: 4

--- Movies & Showtimes ---
1. Ben 10 Alien Force
   1) 10:00 AM
   2) 6:30 PM
2. Odyssey
   1) 1:00 PM
   2) 9:00 PM
3. Spider Man 4
   1) 11:30 AM
   2) 4:15 PM
Select movie number (0 to cancel): 3
You selected: Spider Man 4, Select showtime number (0 to cancel): 2

Seat map for 4:15 PM   ( . = free, X = booked)
   1  2  3  4  5  6  7  8  9 10
A [R]  .  .  .  .  .  .  X  X  .  .
B [R]  .  .  .  .  .  .  .  .  .  .
C [P]  .  .  .  .  .  X  X  .  .  .
D [P]  .  .  .  X  X  .  .  .  .  .
E [V]  .  .  .  .  X  .  .  .  .  .
Legend: [R]=Regular Rs.500   [P]=Premium Rs.750   [V]=VIP Rs.1000
How many seats do you want to cancel? 2

Seat 1 of 2 to cancel:
  Row letter (A-E): C
  Seat number (1-10): 1
  Error: seat C1 is not currently booked.

Seat 2 of 2 to cancel:
  Row letter (A-E): D
  Seat number (1-10): 4
  Seat D4 has been freed and removed from revenue.

1 of 2 requested seat(s) cancelled successfully.

```

```

----- Customer Menu -----
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Back to Main Menu
Enter your choice: 2

--- Movies & Showtimes ---
1. Ben 10 Alien Force
   1) 10:00 AM
   2) 6:30 PM
2. Odyssey
   1) 1:00 PM
   2) 9:00 PM
3. Spider Man 4
   1) 11:30 AM
   2) 4:15 PM
Select movie number (0 to cancel): 3
You selected: Spider Man 4, Select showtime number (0 to cancel): 2

Seat map for 4:15 PM   ( . = free, X = booked)
   1  2  3  4  5  6  7  8  9 10
A [R]  .  .  .  .  .  .  X  X  .  .
B [R]  .  .  .  .  .  .  .  .  .  .
C [P]  .  .  .  .  .  X  X  .  .  .
D [P]  .  .  .  X  X  .  .  .  .  .
E [V]  .  .  .  .  X  .  .  .  .  .
Legend: [R]=Regular Rs.500   [P]=Premium Rs.750   [V]=VIP Rs.1000

```


Screenshot 6: A search result for a booking by name or seat number.

```
===== Movie Ticket Booking System =====
1. Customer Menu
2. Admin Menu
3. Exit
Enter your choice: 2
Admin username (or press Enter to cancel and go back): admin1
Admin password: 1234

Login successful.

----- Admin Menu -----
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Back to Main Menu
Enter your choice: 5
Search by:
  1. Customer name
  2. Seat number
Choose option: 1
Enter customer name to search: Daham
  Daham          | Spider Man 4   | 4:15 PM | Seat C6 | Rs. 750.00
  Daham          | Spider Man 4   | 4:15 PM | Seat C7 | Rs. 750.00
```

Screenshot 7: The revenue report screen, showing totals updating correctly.

```
===== Movie Ticket Booking System =====
1. Customer Menu
2. Admin Menu
3. Exit
Enter your choice: 2
Admin username: admin1
Admin password: 1234

Login successful.

----- Admin Menu -----
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Back to Main Menu
Enter your choice: 6

===== Revenue Report =====
Ben 10 Alien Force
  10:00 AM | Tickets sold: 0 | Revenue: Rs. 0.00
  6:30 PM | Tickets sold: 0 | Revenue: Rs. 0.00
Odyssey
  1:00 PM | Tickets sold: 0 | Revenue: Rs. 0.00
  9:00 PM | Tickets sold: 0 | Revenue: Rs. 0.00
Spider Man 4
  11:30 AM | Tickets sold: 0 | Revenue: Rs. 0.00
  4:15 PM | Tickets sold: 0 | Revenue: Rs. 0.00
-----
Total tickets sold: 0
Total revenue:      Rs. 0.00
```

After booking:

```
----- Admin Menu -----
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Back to Main Menu
Enter your choice: 6

===== Revenue Report =====
Ben 10 Alien Force
  10:00 AM | Tickets sold: 1 | Revenue: Rs. 750.00
  6:30 PM | Tickets sold: 3 | Revenue: Rs. 1800.00
Odyssey
  1:00 PM | Tickets sold: 5 | Revenue: Rs. 2925.00
  9:00 PM | Tickets sold: 0 | Revenue: Rs. 0.00
Spider Man 4
  11:30 AM | Tickets sold: 2 | Revenue: Rs. 1200.00
  4:15 PM | Tickets sold: 6 | Revenue: Rs. 4075.00
-----
Total tickets sold: 17
Total revenue:      Rs. 10750.00
```

Challenges & How They Were Solved

1. Revenue Not Updating on Cancellation

Problem: an early version of `cancelSeat()` cleared the seat's `isBooked` flag without subtracting its `pricePaid` from the showtime's `totalRevenue`, so cancelled seats stayed counted in the revenue report.

Solution: the fix subtracts `pricePaid` and decrements `ticketsSold` before clearing the seat's fields, and a book-then-cancel regression check was added to the team's manual test list. Lesson: the refund calculation must happen before the data it depends on is reset.

2. Crash Prone Input Handling

Problem: an early version read menu choices directly with `scanf("%d", ...)`, which left the input stream in an inconsistent state whenever a letter was typed instead of a number, causing the same prompt to loop without the program ever making progress.

Solution: all input was centralised into `input.c`, reading a full line with `fgets()` and parsing it with `sscanf()`, so a bad line is simply discarded and re-prompted.

3. Merge Conflicts on Shared Headers

Problem: `config.h` and `movie.h` are depended on by almost every module, so two members editing them at once produced conflicts that were riskier to resolve than an ordinary `.c` file conflict, and one early conflict was resolved carelessly and briefly broke the build.

Solution: the team agreed to announce any change to a shared header before pushing, and to resolve conflicts in those files together rather than individually.

GitHub Collaboration Summary

Screenshot : Repository commit history .

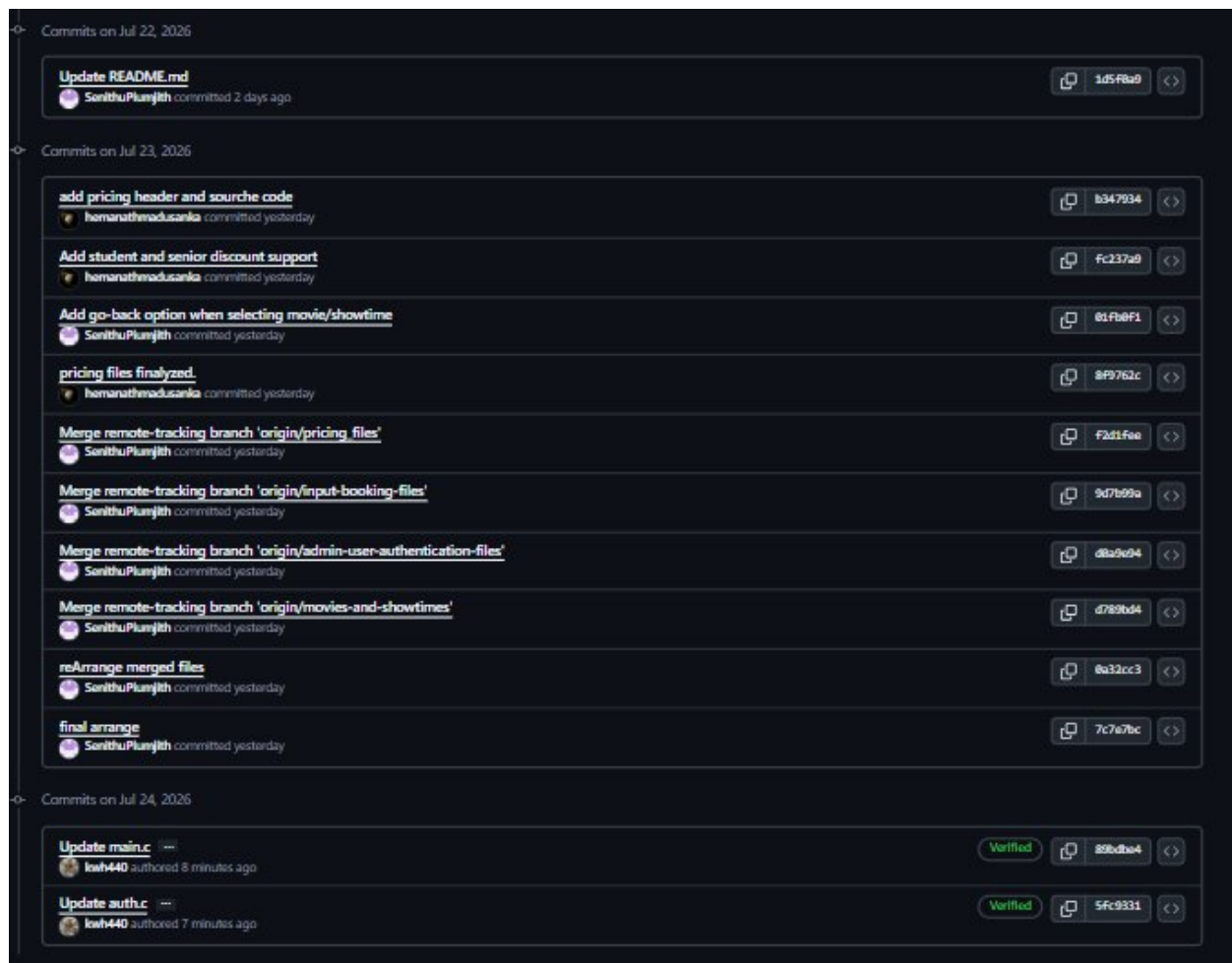
The screenshot displays the commit history of a GitHub repository. At the top, it shows 28 commits and 32 files changed, contributed by 5 contributors. The commits are grouped by date: July 18, 2026, and July 19, 2026.

Commits on Jul 18, 2026

- Add C code for movies and showtimes** by jayashankar094, committed last week. Commit hash: a8d13eb.

Commits on Jul 19, 2026

- Add admin files** by ksh440, committed 5 days ago. Commit hash: 372183a.
- Add admin files** by ksh440, committed 5 days ago. Commit hash: cfc89e4.
- Add user & authentication files** by ksh440, committed 5 days ago. Commit hash: 0fba8a8.
- Add main.c file** by SenithuPlunjith, committed 5 days ago. Commit hash: a951aeb.
- Remove duplicate main.c file** by SenithuPlunjith, committed 5 days ago. Commit hash: ah2f251.
- Update README.md** by SenithuPlunjith, authored 5 days ago. Commit hash: d1eF58a. This commit is marked as **Verified**.
- add seat and display files** by GANETHRAHESHAN, committed 5 days ago. Commit hash: 1d15f53.
- modify display.c with three menu printing functions** by SenithuPlunjith, authored 5 days ago. Commit hash: 04e8be3. This commit is marked as **Verified**.
- Update display.h** by SenithuPlunjith, authored 5 days ago. Commit hash: f88dc64. This commit is marked as **Verified**.
- Add new dedicated login module** by SenithuPlunjith, committed 5 days ago. Commit hash: e449899.
- Add auth.h file** by SenithuPlunjith, committed 5 days ago. Commit hash: 81dF9e2.
- Update main.c** by SenithuPlunjith, authored 5 days ago. Commit hash: 00Fd222. This commit is marked as **Verified**.
- Add config.h file** by SenithuPlunjith, committed 5 days ago. Commit hash: 943a8b8.
- Resolve merge conflict** by SenithuPlunjith, committed 5 days ago. Commit hash: 79e16d8.



The team used a shared GitHub and each member cloned the repository, worked on short lived feature branches (one branch per module or significant change), and pushed commits. Pull requests were used to merge features branch into main. A lightweight issues list tracked outstanding bugs and pending features and This workflow kept main build able for most of development and is one the team intends to reuse in future group projects.